

3axis Robotic Arm

Plastic Gear GEL v1

Three-axis robotic arm controlled by three potentiometers (extension and gripper opening/closing) and two push-buttons (rotation).

Componenti

- 3 MG90S servo (180)
- 6 M2 × 8mm
- 2 M2.5 × 4mm flat screw
- 1 DC geared motor
- L298N motor driver
- Arduino UNO R3
- 3 potentiometer
- 2 button
- Battery 5V
- Battery 9V

The rotating base consists of a bevel gear system (1:3) powered by a DC motor connected to a driver that controls its direction and speed of rotation.

The arm comprises two joints and a gripper (spur gear mechanism), all operated by MG90S servos.

The servomotors are powered by a 5V supply, while a 9V supply powers the motor driver

ASSEMBLY

M2 screws were used for assembly to secure the servomotors, and M2.5 screws to attach the “limbs” to the servomotors, while the gripper component connects via a snap-fit mechanism

CODE

The code uses the `map()` function to convert potentiometer values (0–1024) into angles for the servomotors (0–180)

As for the pushbuttons, to simplify wiring and avoid adding a 10kΩ resistor, they are connected directly to GND and a GPIO pin and configured as `INPUT_PULLUP` this way, the `digitalRead()` function returns 1 when the button is not pressed (and vice versa)

3d model i partially used :

<https://makerworld.com/it/models/1116022-robot-arm-claw-for-mg90s-micro-servo-motor#profileId-1113520>

i took the spur gear gripper arm and adapted the hole

<https://makerworld.com/it/models/2697691-tt-motor-mount-version-2>

i used it to put the DC motor at a fixed distance for the bevel gear to work

<https://makerworld.com/it/models/439754-stemfie-power-transmission-hub-pln-idx-fxd-ttmotor#profileId-345381>

used it as an adpter for the motor